

Domain Notes

Overview

This dataset models paid social ad performance: campaigns, individual ads, the users who see them, and the events generated when users interact with ads. It supports analysis of reach, engagement, conversion efficiency, and spend, structured as a star schema for BI reporting.

Table Roles

- **Events table (fact table):** the core log of every interaction (impressions, clicks, shares, comments, purchases). Every KPI is derived from this table.
- **Ads table (dimension):** metadata about each ad, including which platform it ran on, its creative format, and who it targeted.
- **Campaigns table (dimension):** the strategic layer above ads: budget, timeframe, and campaign name, used for spend-based metrics.
- **Users table (dimension):** demographic and interest data on the people who interacted with ads, used to check whether targeting matched actual audience behavior.

Field Reference: Events

Field	Description	Analytical use
event_id	Unique row identifier	Primary key
ad_id	Links to the ad shown	Join to ads table
user_id	Links to the user who acted	Join to users table
timestamp	Exact moment of the event	Builds day/week/month hierarchy
day_of_week	Derived weekday label	Weekday vs. weekend comparison
time_of_day	Derived time bucket	Identify peak engagement windows
event_type	Impression, Click, Share, Comment, or Purchase	Funnel-stage analysis

Field Reference: Ads

Field	Description	Analytical use
ad_id	Unique ad identifier	Primary key
campaign_id	Parent campaign	Join to campaigns
ad_platform	Platform the ad ran on	Platform comparison
ad_type	Creative format	Format comparison
target_gender	Intended gender audience	Targeting accuracy check
target_age_group	Intended age audience	Targeting accuracy check

Field	Description	Analytical use
target_interests	Intended interest categories	Match against actual user interests

Field Reference: Campaigns

Field	Description	Analytical use
campaign_id	Unique campaign identifier	Primary key
name	Campaign label	Reporting/filtering
start_date / end_date	Campaign window	Duration and pacing analysis
duration_days	Derived length	Pacing comparison
total_budget	Spend allocated	Basis for cost-efficiency metrics

Field Reference: Users

Field	Description	Analytical use
user_id	Unique user identifier	Primary key
user_gender / user_age / age_group	Demographics	Segment-level performance
country / location	Geography	Geographic reach analysis
interests	Stated interests	Compare against ad targeting

How It Fits Together

Events link out to ads, users, and (through ads) campaigns, giving a fact table surrounded by dimension tables, which is the standard star-schema pattern for this kind of BI model.